

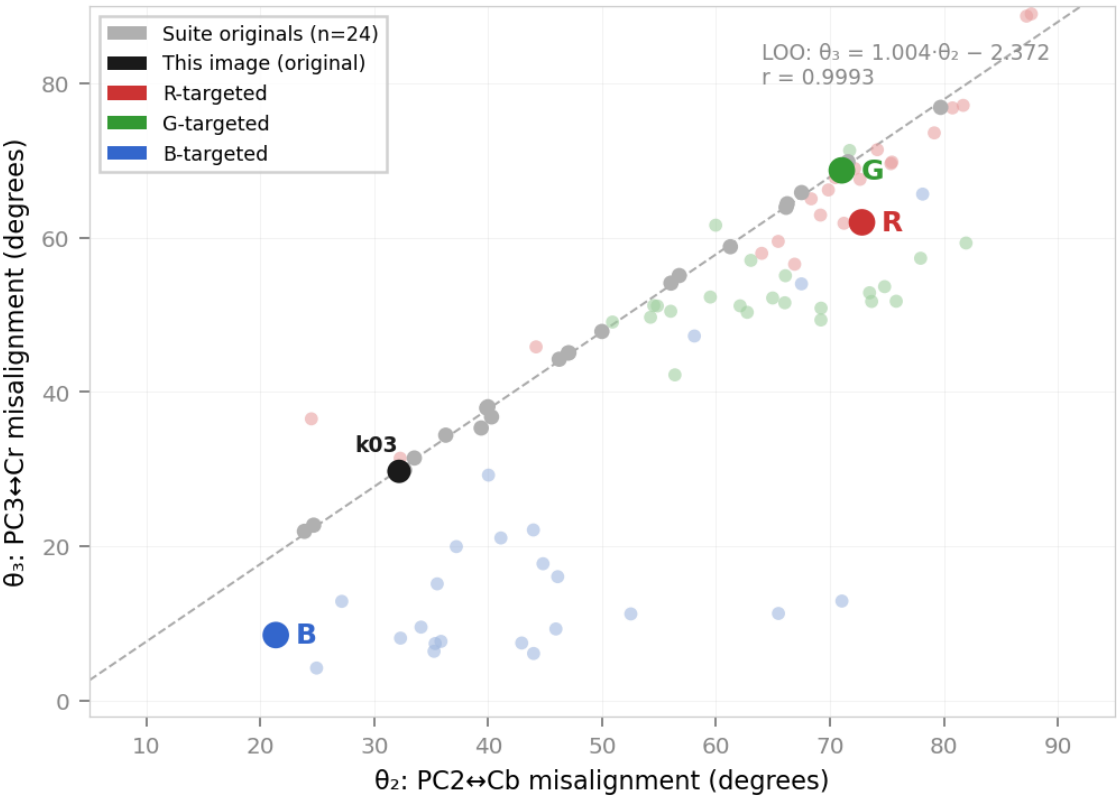
Compression Profile: KODIM03

768x512 | 24-bit

RGB | PNG (lossless)

hats | Three-Dimensional (PC1 = 69.27%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.7185	0.2890	0.5534	0.520	32.17°	29.72°	-0.21°
R-targeted	-0.0562	-0.1320	0.9526	0.380	72.81°	62.01°	-8.72°
G-targeted	-0.1047	0.9468	0.0800	0.377	71.04°	68.75°	-0.20°
B-targeted	0.9015	0.1506	0.1243	0.392	21.35°	8.49°	-10.57°

R-targeted isolates red (R-G, R-B < 0.13) while preserving G-B coupling at 0.953.
G-targeted isolates green (R-G, G-B < 0.10) while preserving R-B at 0.947.
B-targeted preserves R-G at 0.901 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold (r = 0.9993).

Compression Profile: KODIM03768x512 | 24-bit
RGB | PNG (lossless)
hats | Three-Dimensional (PC1 = 69.27%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

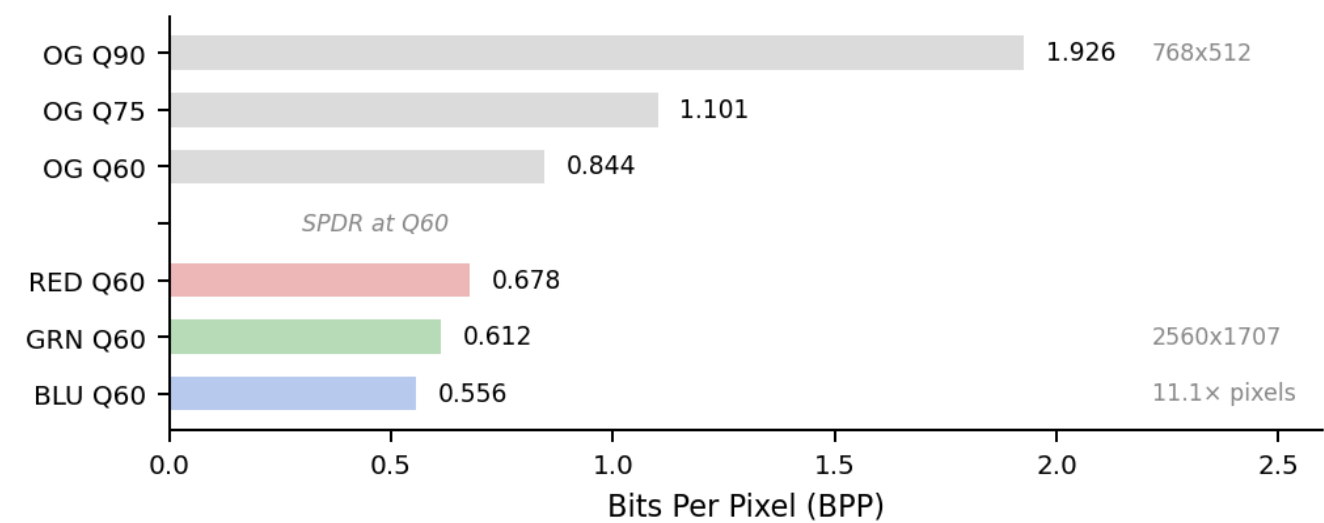
ORIGINAL (kodim03)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	491.1 KB	10.231	393,216
JPEG Q90	JPEG	768x512	92.4 KB	1.926	393,216
JPEG Q75	JPEG	768x512	52.8 KB	1.101	393,216
JPEG Q60	JPEG	768x512	40.5 KB	0.844	393,216

REDISTRIBUTED (kodim03 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	361.8 KB	0.678	4,369,920
GRN Q60	JPEG	2560x1707	326.6 KB	0.612	4,369,920
BLU Q60	JPEG	2560x1707	296.8 KB	0.556	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1x the original pixel count.

Compression Profile: KODIM03

RGB | PNG (lossless)

hats | Three-Dimensional (PC1 = 69.27%)

768x512 | 24-bit

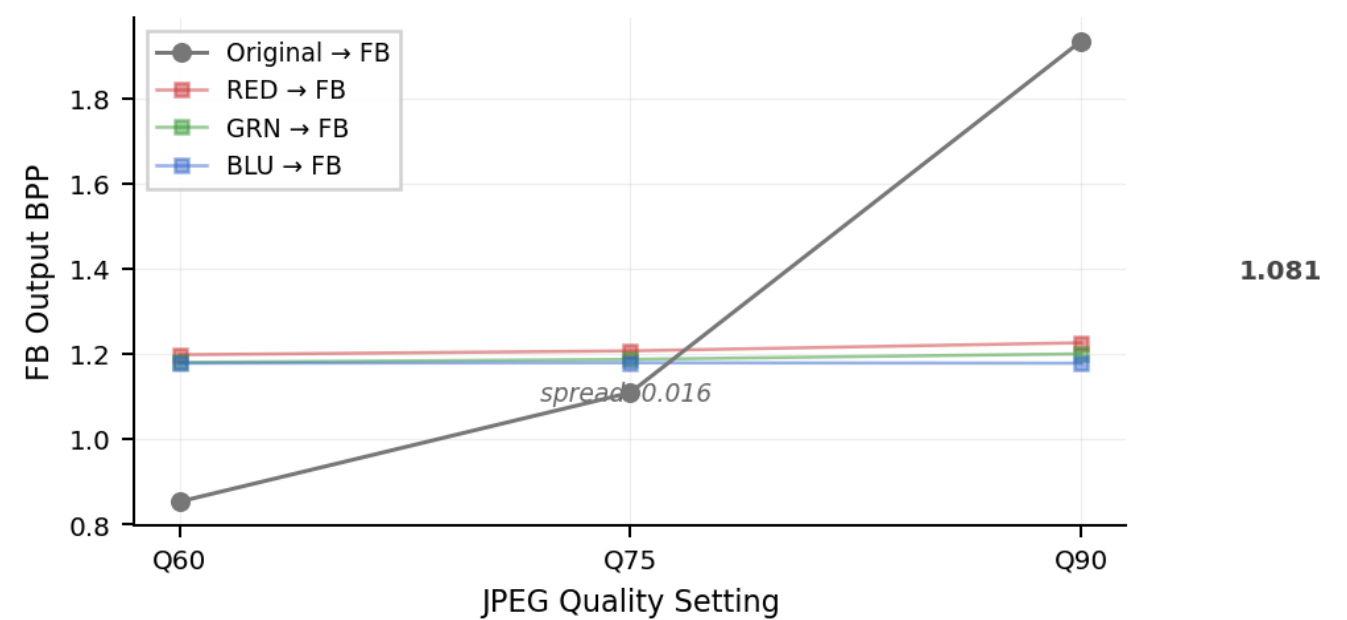
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	1.792	—
R-targeted → FB	2560x1707	2048x1366	1.227	31.5%
G-targeted → FB	2560x1707	2048x1366	1.201	33.0%
B-targeted → FB	2560x1707	2048x1366	1.179	34.2%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.013	43.5%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.5173 vs 0.520 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.081 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.016 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim03

Best BPP (Q60, direct)	0.556	2560x1707, 4,369,920 px
Best BPP through FB	1.179	2048x1366 output
vs OG Q90 → FB	39.1%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216